

BIM254: Course project – Ashwin Amalaruban

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Alkan et al.'s "CRISPR-Cas9 off-targeting assessment with nucleic acid duplex energy parameters" introduces two prediction models, **CRISPROff** and **CRISPRspec** that utilises energy parameters for RNA-RNA, DNA-DNA and RNA-DNA duplexes¹. CRISPROff scores the energy for Cas9-gRNA-DNA complex for particular off-target binding and CRISPRspec gives a specificity score for a guided RNA using free energies of all possible bindings. CRISPROff performs much better than other off-target scoring methods with a low False Positive Rate (FPR) and a high True Positive Rate (TPR). On the other hand, specificity score of CRISPRspec agrees well with experimental results with a strong negative correlation between the predicted specificity and off-target read counts.

Idea #1: Human genome was the primary focus for benchmarking studies in this paper. Mouse and rat genome from Anderson2018 dataset² can be used and compared with the results seen in Fig 2,3 and 5.

Idea #2: Effect of other CRISPR proteins such as Cas13³ can be studied with minor modifications in the energy model. Kinetics of various gRNA-DNA bindings are to be measured from biochemical profiling datasets for this purpose.

Either of this idea could be implemented along with a literature review on further developments with respect to this work^{4,5,6}.

References

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